YI DENG

(+1) 434-351-1412 | yideng125@gmail.com | Charlottesville, VA, USA

EDUCATION

University of Virginia Aug. 2025 – Dec. 2026
M.E. in Computer Engineering | GPA: 3.70 / 4.00

- Relevant coursework: Wireless Sensing, Machine Learning, Human-Computer Interaction (HCI).

Shenzhen MSU-BIT University Aug. 2021 – Jun. 2025
B.Eng. in Electronic & Computer Engineering | GPA: 3.71 / 4.00

- Core courses: Modern Communication Principles, Database Principles, Machine Learning.

PUBLICATIONS

First-authored

[1] **Y. Deng**, et al. "Cloud–Edge EEG Emotion Recognition under Communication Constraints," *IEEE International Conference on Cloud Computing Technology and Science (CloudCom)*, 2025.

[2] **Y. Deng**, et al. "Reviving Affective Signals: A Generative Data Reconstruction Framework for EEG Digital Twins under Communication Constraints," *IEEE Open Journal of the Communications Society*.

[3] **Y. Deng**, et al. "The Influence of Mamba Model Hyper-parameters on Training Time and Accuracy," *IEEE International Conference on Smart Internet of Things (SmartIoT)*, 2024.

Co-authored

[4] M. Li, **Y. Deng**, et al. "Personalized Real-Time Epileptic Seizure Alert on Edge Devices via Few-Shot Meta-Learning," *IEEE International Conference on Cloud Computing Technology and Science (CloudCom)*, 2025.

[5] M. Li, **Y. Deng**, et al. "GenDT-MAML: A Data-Efficient Framework for Personalized Seizure Prediction Using Generative Digital Twins and MAML," *IEEE Open Journal of the Communications Society*.

RESEARCH EXPERIENCE

Multimodal Physiological Sensing in Driving Experiments Mar. 2026 – Aug. 2026
Research Assistant, University of Louisville

- Built preprocessing and quality-control pipelines for 16-channel EEG, three-wavelength PPG, and EDA across ~90 driving trials, covering filtering, artifact screening, channel interpolation, peak-quality checks, and multimodal temporal alignment.
- Characterized motion artifacts, missing or low-dynamic signals, and inter-participant variability that limit event-level modeling, and established trial- and event-level quality criteria for defensible analysis.
- Extracted participant-specific baseline and event-stage features (heart rate, frontal theta, occipital alpha, skin conductance) and compared them across driving-event stages.

Cloud–Edge EEG Emotion Recognition under Communication Constraints Nov. 2024 – Dec. 2025
First-authored: CloudCom 2025, IEEE OJ-COMS

- Designed and implemented an end-to-end EEG emotion recognition pipeline using CNN–LSTM–Attention architectures in PyTorch, processing multi-channel EEG signals from preprocessing through classification.
- Modeled cloud–edge deployment by simulating realistic communication constraints—packet loss, lossy compression, and bandwidth limits—to characterize how transmission degradation impacts downstream emotion classification accuracy.
- Developed generative reconstruction modules based on GAN and diffusion models to recover degraded EEG segments at the cloud side, restoring downstream inference accuracy under constrained channels.
- Conducted system-level analysis of accuracy, latency, and transmission cost trade-offs across deployment configurations, informing design guidelines for edge-deployed biosignal inference.

Performance Analysis of Mamba State-Space Models Apr. 2024 – Jul. 2024
First-authored: SmartIoT 2024

- Designed controlled experiments in a Linux environment to study how key hyperparameters (learning rate, batch size, SSM state expansion factor) affect Mamba training time and downstream accuracy.
- Built an automated benchmarking pipeline evaluating Mamba performance across multiple NLP datasets (SST-2, IMDB, AG News), enabling reproducible cross-dataset comparison.
- Identified performance trade-offs and training-stability regimes under different hyperparameter configurations, providing empirical guidance for deploying state-space models in resource-constrained settings.

OTHER PROJECTS

Contactless Respiration Sensing with WiFi CSI and FMCW Radar Aug. 2025 – Dec. 2025
Team coursework, Wireless Sensing, University of Virginia

- Ran and tested open-source WiFi-CSI and FMCW-radar pipelines for respiration estimation, including preprocessing and frequency-domain analysis across sensing conditions.

Construction and Application of Raman Spectrometer Apr. 2023 – May 2024
Departmental research project, Shenzhen MSU-BIT University

- Built and calibrated a Raman spectrometer and a UV-Vis absorption spectrometer from optical components, ensuring optical-path alignment and spectral accuracy.
- Designed and implemented additional measurement features, completing the engineering department's first in-house Raman spectrometer for educational and research use.

INDUSTRY EXPERIENCE

Fusheng Innovation (Chengdu) Technology Co., Ltd. Apr. 2024 – Sep. 2024
Research and Development Engineer (Intern)

- Designed and implemented backend modules in Go for a trading system, integrating broker APIs for order handling and status updates.
- Maintained relational databases for transactional data and supported testing of API behavior under network latency and failure scenarios.

Shenzhen UBTECH Robotics Corp., Ltd. Nov. 2024 – Feb. 2025
Product Intern

- Validated AI-based obstacle recognition through functional testing and supported system-level evaluation of robotic vacuum products, focusing on sensing, navigation, and edge cases.
- Collected and analyzed user feedback and test results to support iterative product improvements.

TECHNICAL SKILLS

- **Programming:** Python (proficient), C/C++, Go; experience with object-oriented design, algorithm implementation, and performance-aware development.
- **Machine Learning & Deep Learning:** PyTorch; CNN, LSTM, attention, and state-space (Mamba) architectures; generative models (GAN, diffusion); time-series modeling and feature extraction for biosignal learning tasks.
- **Signal Processing & Sensing:** EEG preprocessing and analysis; optical spectroscopy (Raman, UV-Vis); coursework exposure to WiFi CSI and FMCW radar pipelines for contactless physiological sensing.
- **Embedded & Control Systems:** MATLAB / Simulink for system modeling and simulation; basic controller design and tuning (e.g., PID).
- **Networking & Tools:** TCP/IP and HTTP communication, TCP/UDP module development in Python; Linux, SQLite, Git.

HONORS & AWARDS

- University-level Third-class Academic Scholarship, Shenzhen MSU-BIT University, 2022–2023 (Top 30%).
- Third Prize, 15th Blue Bridge Cup National Software and Information Technology Competition, University Group B (C/C++), Guangdong Province, Apr. 2024.